

Agentic AI for Tech Professionals

Trainer: Goh Kheng Yee

TRAINER PROFILE



Trainer Kheng Yee

- HRDC Accredited Trainer
- M.Sc. in Analytics, Georgia Tech, USA
- 6 years of Industrial Experience in AI & Data Science
- Trainings focused on AI, Machine Learning, Python, SQL, Data Analytics, and Data Visualization.

Professional Experience Highlights

Consulting

Led development of AI-powered learner engagement systems for an education institution. Contributed to public sector projects by applying advanced analytical models.

Media

Built AI/machine learning models to identify audience segments, enabling the shift to addressable TV advertising.

Banking

Developed machine learning models for banking products and optimized lead quality through model overlay techniques.

Professional Certificates



HRDC Accredited
Trainer



AWS Certified
Cloud Practitioner



Salesforce Certified
Tableau Desktop
Foundations



Google Cloud
Certified Cloud
Digital Leader

Introduction to Artificial Intelligence

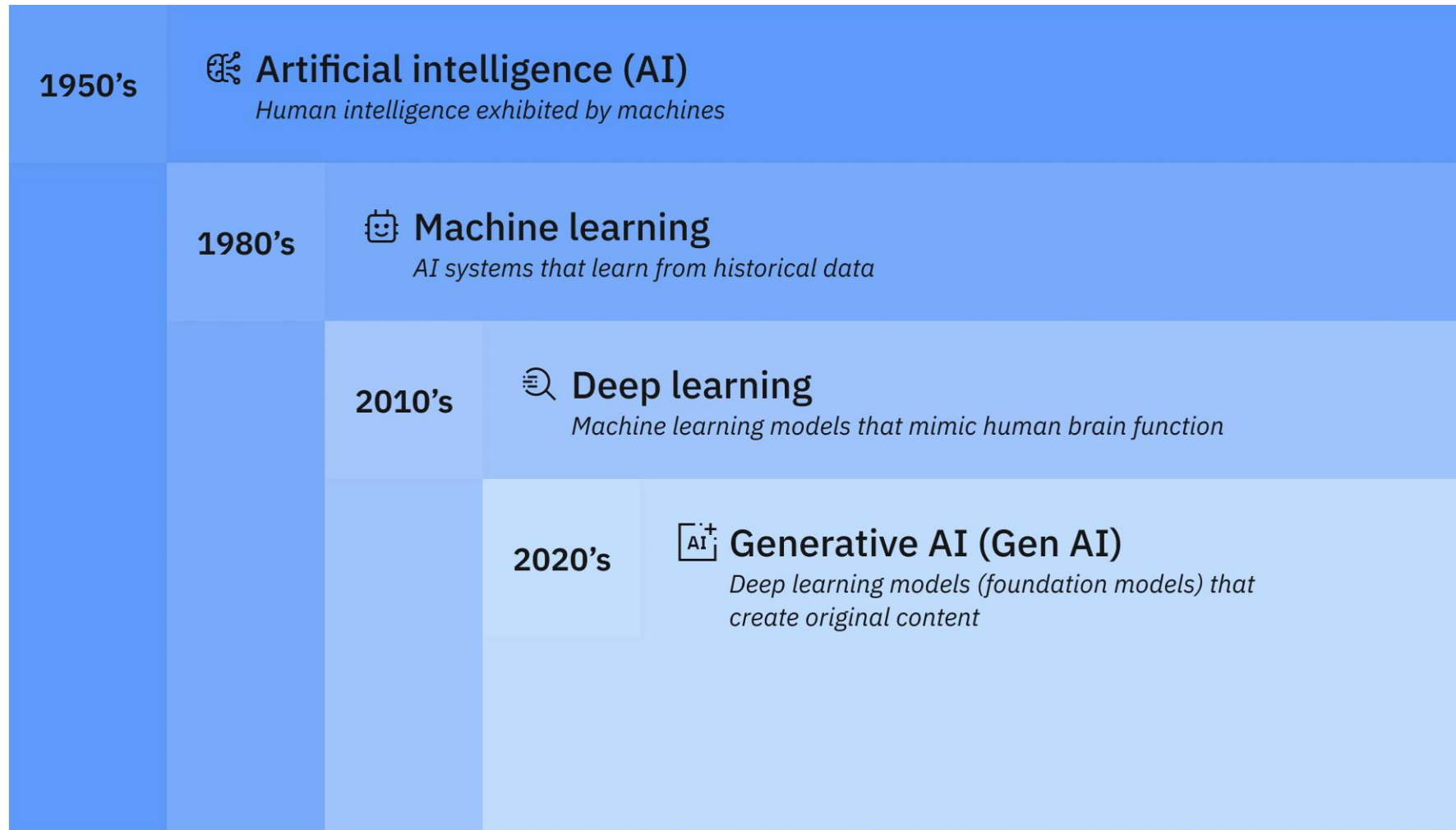
What is AI?

The science of making machines think and act like humans.

Key Concept

AI analyses large amounts of data, finds patterns, and uses those patterns to make predictions.





Emergence of New AI-related Job Roles



AI Agent Specialist

Designs autonomous systems where LLMs use tools to execute multi-step workflows without human intervention.

Core Competencies

LangGraph

Orchestration

Tool Calling



GenAI Engineer

Focuses on integrating Large Language Models into applications using RAG and advanced prompting techniques.

Core Competencies

RAG Pipelines

LangChain

Fine-Tuning



AI Engineer

Builds end-to-end production systems, handling model deployment, scalability, and infrastructure (MLOps).

Core Competencies

Python

Model Serving

Cloud (AWS)



Prompt Engineer

Optimizes model inputs to achieve consistent, high-quality outputs and designs evaluation datasets.

Core Competencies

Context Design

Few-Shotting

Logic Flow

AI in Everyday Life



Popular GenAI Tools



AI vs Traditional Automation

Rules do the same thing every time; AI adapts to what it is given.

Traditional Automation

- Follows fixed rules
- Same steps every time
- Best for repetitive actions
- Breaks on exceptions

AI

- Handles variations
- Adapts to context
- Best for thinking tasks
- Manages exceptions

Copilot plans, products and environment strategy

ASSIST	BUILD AGENTS	AUTOMATE
Copilot Chat Conversational assistance	Agent Builder Focused declarative agents	Power Automate Workflow execution
Microsoft 365 Copilot Work and app context	Copilot Studio Tools and channels	

Designing a conversational bot

BOT PERSONA

Role

Scope & intents

Tone

Escalation

Clear boundaries: what the bot can and cannot do

INSTRUCTION STRUCTURE

01	02	03	04	05	06
Role	Goal	Sources	Steps	Constraints	Output format

What is Copilot Studio?

- A low-code platform by Microsoft that lets you build your own AI agents for business use.
- Build AI agents using drag-and-drop
- Connect to your company data
- Automate tasks and workflows

Copilot Studio



AI-Native Architecture

Built for conversational AI agents



Microsoft 365 Integration

Seamless with Teams, SharePoint



Enterprise Security

RBAC, compliance, data governance



Power Platform

Connect to Power Apps, Power BI

GenAI vs Agentic AI

From answering a single prompt to completing a whole task.

GenAI

One answer at a time

- You give the full context
- Responds to each prompt
- Great for drafts and ideas

AI Agent

Multiple steps until the job is done

- Plans and uses tools
- Works towards a goal
- Great for workflows

AI Agents vs. Traditional Prompt-based AI

Understanding the shift from reactive tools to proactive autonomous assistants.



Prompt-based AI



The Analogy

Asking a Colleague a Question

"Hey, what is the capital of France?"



Interaction Model

One-time Q&A

You provide full context. It gives one answer.



Behavior

Reactive

It waits for your specific command to act.



Outcome

Single Text Response

Great for drafts, definitions, and summaries.

VS



AI Agents



The Analogy

Hiring an Assistant for a Role

"Plan my travel to Paris and book the hotels."



Interaction Model

Multi-step Autonomous Work

It plans steps, uses tools, and iterates on goals.



Behavior

Proactive

Can ask clarifying questions and take initiative.



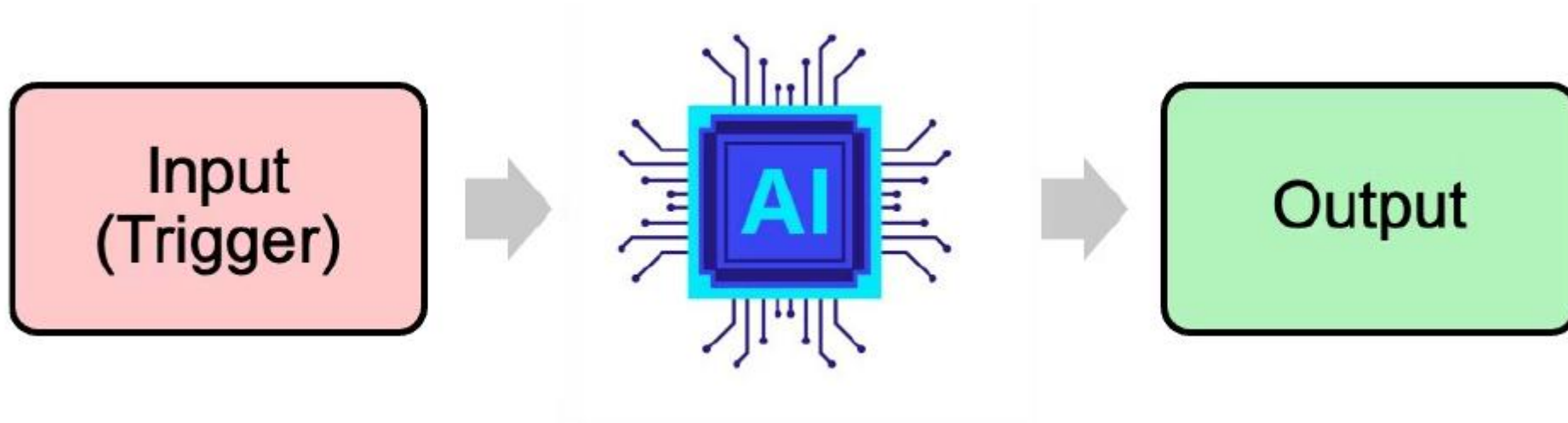
Outcome

Complete Project Ownership

Ideal for research, workflows, and monitoring.

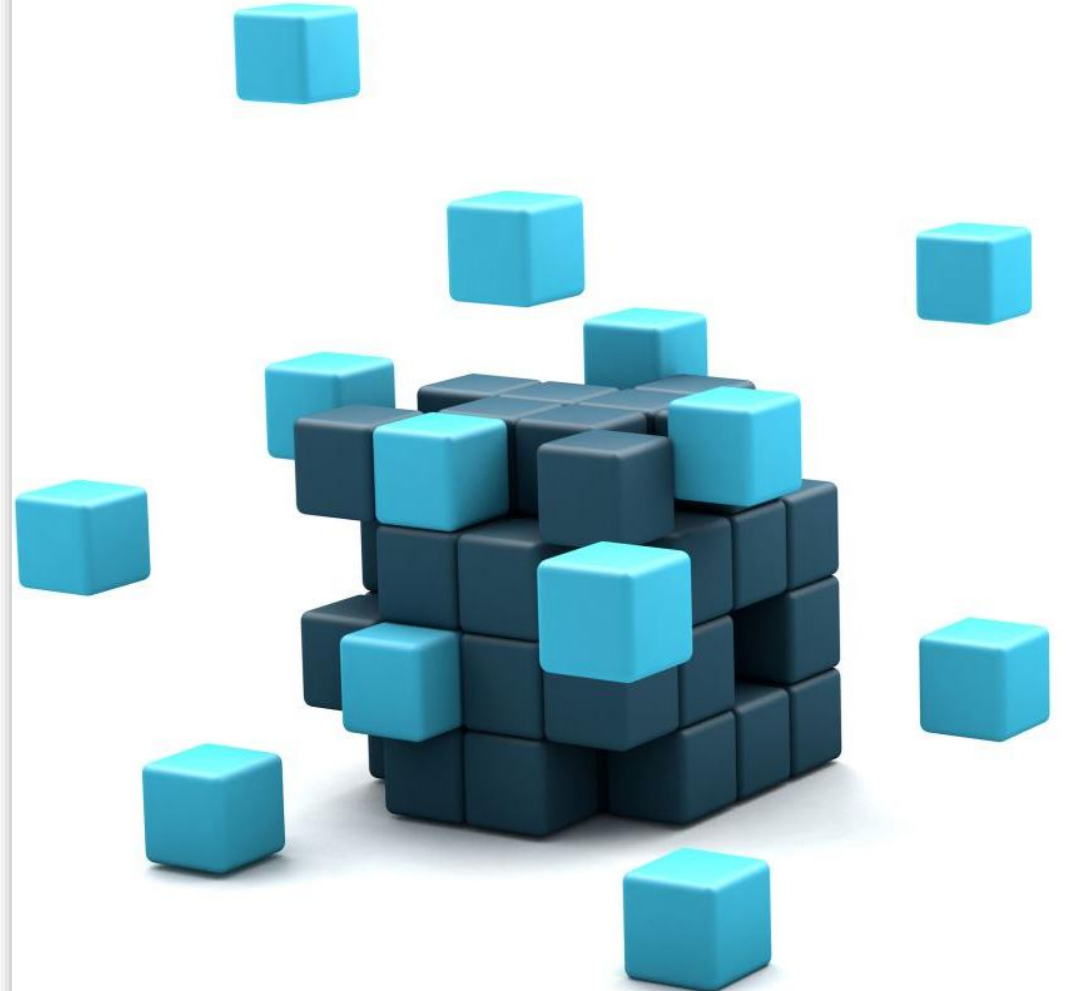
Takeaway: While prompt-based AI helps you *do a task* faster, AI Agents help you *achieve an outcome* by managing the process for you.

Workflow



Building Blocks of an AI Agent

- Goals
- Rules
- Memory
- Tools
- Data



Building Your First Agent

Move from theory to practice by standing up, testing, and publishing a working agent.

1 Creating an agent from scratch

2 Understanding agent harnesses (standard vs. GitHub Copilot harness)

3 Setting agent instructions and personality

4 Testing agents in the test panel

5 Publishing your first agent

Integration Flow

1

Configure LLM

Set up model, temperature, and token limits

2

Define System Prompts

Create role-based instructions for the agent

3

Add Actions

Configure tool calling for external APIs

4

Test & Validate

Verify structured output and error handling

AI rules and deterministic workflow rules

AI guidance

Interpret intent, ask clarifying questions and draft an explanation. Model output can vary.

Deterministic controls

Validate types, permissions and approved tolerances. Enforce approval gates before impactful actions.

Knowledge grounding with SharePoint and Excel



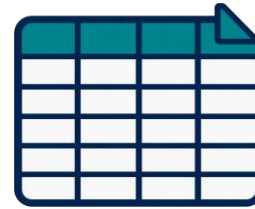
SharePoint

Controlled knowledge

Named owner + clear titles

Metadata + version history

Permissions + review



Excel

Structured records

Structured tables

Stable columns + data types

Unique IDs + controlled values

An agent that invokes a workflow

01

Define

Name, description, instructions and starter prompts establish the agent's scope.

02

Connect

Add an agent flow or as a tool. Describe when to call it.

03

Validate

Use typed inputs and outputs. Check fields, permission, duplicates and approvals.

04

Return

Report a verified outcome and reference ID.

Combining SharePoint, Excel and SQL data

SharePoint

Controlled procedures

Retrieve and cite the applicable version. Keep knowledge retrieval separate from actions.

Excel

Technician or batch records

Read structured tables. Validate row identity and control duplicate or concurrent updates.

SQL

Authoritative production history

Query by permitted identifiers. Apply appropriate authentication and return only needed fields.

Scheduled collection and summarization

01

Trigger

Use recurrence for a shift digest, an event for a new failure, or conversation for a user request.

02

Collect

Fetch new or failed records. Validate completeness and calculate operational indicators.

03

Summarize

Use bounded counts, exceptions, owners and source links. Expose missing records.

04

Deliver

Send approved output to Teams or email. Retain the execution log and notification status.

Engineering test and production solution



Technician

Capture the test

Chat + confirm values

Tool writes to Excel

Receive record ID



Supervisor

Review the failure

Check failed test + tolerance

Resolve evidence gaps

Approve escalation



Production

Assess the impact

Enrich from SQL

Check equipment + work order

Summarize the shift

Testing, governance and SKILL.md

A skill is a reusable capability for a specific task

WHAT IT CONTAINS

Name + description

Markdown instructions

Optional scripts, references, templates
and other resources

WHY USE SKILLS?

Reusable	Use across agents
Modular	Keep tasks focused
Shareable	Export files or packages
Clear	Understand and maintain behavior

Testing, governance and SKILL.md

project-status-dashboard/

SKILL.md

Required: when to use the skill + main instructions

references/

Knowledge the agent reads

scripts/

Code the agent runs

assets/

Files the agent uses in its output

Course Evaluation

- <https://feedback.iverson.com.my/ifs/>
- Class ID: 58911

